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Quantifying the convection heat transfer coefficients in the molten metal jetting process with experiments, theoretical analysis, and computational modeling

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Outline

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Highlights

- Convection coefficients are quantified for the molten metal jetting process.
- Print platform motions during printing exhibit enhance convective heat transfer.
- Argon flow and high-temperature printhead proximity reduce convective heat transfer.
- Accurate convection coefficient values promise accurate thermal simulations results.

Abstract

Molten metal jetting (MMJ) is an emerging additive manufacturing process that produces high-resolution metal parts from cost-effective feedstock, making it attractive for industrial applications requiring intricate components. However, modeling the thermal behavior of MMJ builds remains a challenge because a suitable convective heat transfer coefficient governing part cooling is unknown. An accurate MMJ convection coefficient is needed to enable thermal simulations that predict cooling rates and the resulting microstructure during process parameter optimization studies.

This study quantified MMJ convective heat transfer coefficients under four cooling scenarios: natural convection in air; forced argon flow with heated printhead proximity; print platform motion; and all three conditions combined. Convective coefficients under each condition were evaluated using a novel adaptation of fin theory together with Ansys Fluent computational fluid dynamics (CFD) simulation and experimental temperature measurements. To increase accuracy of the classic fin theory model, location-dependent ambient temperature values were used instead of assuming a fixed global ambient temperature. This adaptation captures the specific nature of the MMJ process in which ambient temperature values vary greatly as a function of proximity to the high-temperature molten metal droplet jetting assembly that is positioned above the part surface during printing.

Both theoretical analysis (the modified fin theory model) and CFD results indicated that platform motion enhanced convective cooling, while hot argon flow and proximity to the heated print head contributed to localized heating. For the most representative MMJ printing condition involving a moving platform, close proximity of the molten metal jetting nozzle, and heated argon flow, a constant convection coefficient of approximately $14\text{W/m}^2\cdot^\circ\text{C}$ was found to be most suitable for use in molten metal jetting thermal simulations.

Keywords

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgements

Appendix A. Analytical Correction Applied...

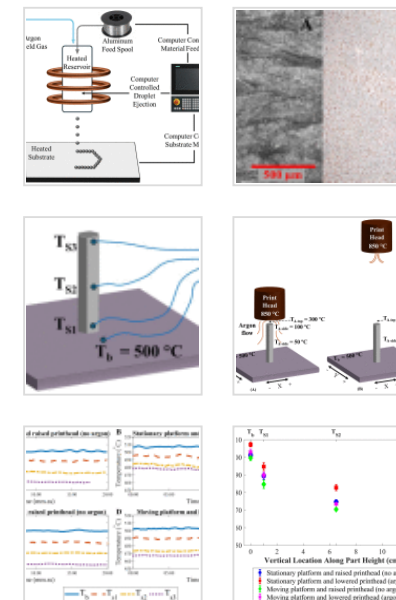
Appendix B. Dimensionless numbers and ...

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Molten metal jetting; Additive manufacturing; Thermal modeling; Convection coefficient; Fin theory; Theoretical analysis; Computational modeling

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Nomenclature

A_c

Cross-sectional area of the part

h, h_{conv}

Convective heat transfer coefficient

h_{top}

Convection coefficient for part's top surface

h_{side}

Convection coefficient for part's side walls

k

Thermal conductivity

L

Height of the part

P

Perimeter of the part geometry

$S1, S2, \text{ and } S3$

Thermocouple locations on the printed part for temperature measurements during the experiments.

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[Table 2. Mesh density and the resulting t...](#)

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T_{∞}

Surrounding temperature

 T_b

Print platform temperature

 T_{A-top}

Ambient temperature for part's top surface

 T_{S1} , T_{S2} , and T_{S3}

Temperature recorded at locations S1, S2, and S3

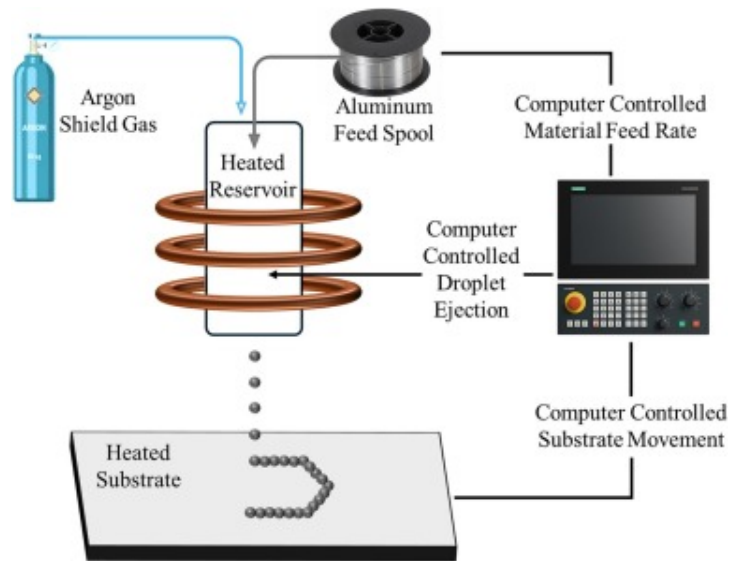
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Distance on part's height measured from the print platform

1. Introduction

Metal additive manufacturing (MAM) has emerged as a transformative technology with extensive applications across aerospace, automotive, healthcare, and consumer products [1], [2]. Among various additive manufacturing processes, molten metal jetting (MMJ) has garnered significant attention for its ability to fabricate intricate, high-quality metal parts using cost-effective feedstock materials [3], [4], [5], [6]. Molten metal jetting (Fig. 1) builds parts layer by layer, following a toolpath created from a CAD model, by jetting molten metal droplets on demand onto a substrate to produce the desired 3D part geometry [7], [8]. This process uses induction or resistance heating to maintain the molten metal reservoir above the liquidus temperature of the metal. A jetting mechanism, powered by either magnetohydrodynamic [7], [9], [10] or piezoelectric [5] actuators, ejects droplets toward the substrate. Continuous argon flow around the printhead nozzle prevents oxidation of the molten metal droplets [11], [12]. Upon deposition, the molten metal droplets retain sufficient thermal energy to bond with the previous layer, followed by rapid solidification, ensuring structural integrity and mechanical strength [5]. Post-

processing steps such as machining, surface finishing, and heat treatment may further enhance the mechanical strength and surface quality of fabricated parts.



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Fig. 1. General schematic of the MMJ process.

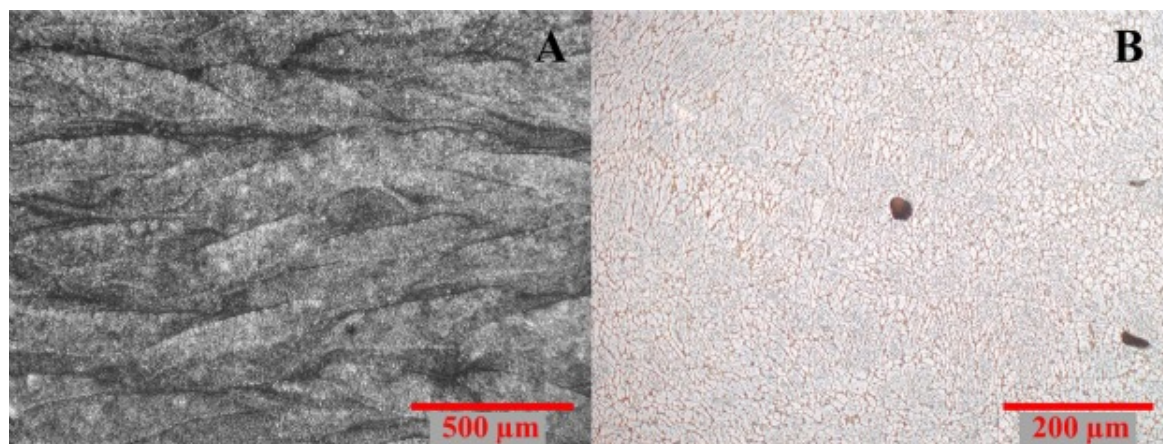
Molten metal jetting has the potential to process a broad range of metal alloys, offering material versatility for various industries [13], [14], [15]. Unlike powder-based processes, molten metal jetting utilizes wire or rod feedstock, which reduces raw material costs. Additionally, molten metal jetting minimizes waste by depositing material only where it is needed, making it well-suited for both prototyping and small- to medium-volume production [16]. This process enables the fabrication of geometrically complex parts that are difficult to produce with traditional subtractive manufacturing or casting processes. Despite its promise, molten metal jetting faces technical challenges, including droplet temperature control, viscosity stability, and reliable droplet ejection [8]. Structural defects like porosity, due to poor droplet spreading and surface roughness, can compromise mechanical properties [17], [18], [19]. These process limitations present opportunities for

engineering and scientific advancements, enabling broader adoption of molten metal jetting in safety-critical industries such as aerospace and biomedical engineering, where precision and mechanical reliability are paramount [20]. Addressing these challenges requires an in-depth understanding of the thermal-fluid interactions and process-structure-property relationships inherent in molten metal jetting [21].

Conducting statistical experiments to investigate process-structure-property relationships is both energy-intensive and time-consuming due to the large set of process parameters involved. Moreover, the dependence of process parameters on the starting material and part geometry limits the transferability of knowledge from one material or design to another. Therefore, modeling the process and simulating the effects of molten metal jetting parameters on part quality is both essential and urgent. A critical barrier to developing accurate modeling for molten metal jetting is the limited understanding of the convection coefficient—a key parameter governing heat transfer between the fabricated part and its surrounding environment. Despite advancements in computational modeling techniques, the convection coefficient for molten metal jetting has yet to be comprehensively studied or quantified. This uncertainty impedes accurate thermal simulations and indirectly contributes to process instabilities. Addressing this gap could enable precise process control and reduce defects in critical metal additive manufacturing applications.

By understanding and controlling the thermal dynamics through simulations, researchers can identify the process parameters needed to mitigate defects such as porosity and surface roughness, leading to improved part quality and broader application of the molten metal jetting process in critical industries. In Fig. 2 (A), weak interlayer bonding is indicated by distinct boundaries between adjacent layers. Similarly, porosity (Fig. 2, B) can result from either shielding gas entrapment or excessive energy creating deep melt pools causing near-spherical voids at the molten pool base [22]. Parts with these defects fail to meet standard testing requirements in the biomedical, automotive, and aerospace industries. Given the critical role of heat transfer in determining part quality, thermal simulations become indispensable for optimizing process parameters and minimizing

defects. For accurate simulation results, precise heat transfer coefficients must be incorporated into the thermal calculations.



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Fig. 2. Optical micrographs of material printed by molten metal jetting process, showing droplet boundaries (A) and porosity (B).

Many prior molten metal jetting simulations were limited to one-dimensional (single droplet or pillar) [22], [23], [24], [25], [26] and two-dimensional (thin-wall) models [27], [28]. These studies revealed droplet impact, thermal distribution, and solidification behaviors [29] but often neglected convective heat transfer to reduce computational complexity [30], [31], [32]. By ignoring convection, researchers could isolate droplet deposition phenomena such as impingement and spreading [33], [34], [35], [36]. However, these 1D and 2D models cannot capture the true heat transfer environment of a full 3D build. Some simulations have been extended to small-scale three-dimensional regions of the build [37], [38], providing detailed local temperature profiles but still failing to capture thermal interactions across an entire part [39]. Moreover, like the 1D/2D cases, these 3D models either omitted convection entirely [27] or considered only natural convective cooling [22], [37], neither of which reflects the dynamic environment of the

molten metal jetting process. Thus, while these studies provide valuable insights, they are only applicable to microscale thermal predictions and cannot be extended to macroscale properties of functional parts fabricated with molten metal jetting.

To address the gaps in molten metal jetting simulations, researchers can benefit from examining thermal simulations of large parts fabricated using directed energy deposition (DED). Since both processes operate in an open environment, comparisons between them are relevant. In directed energy deposition simulations, convection coefficients are critical for accurate thermal modeling [40], [41]. These coefficients influence the thermal profiles, cooling rates, and predicted temperature distributions, which are essential for ensuring part quality. Reports on directed energy deposition simulations often use convection coefficients that are either empirically determined [42] or approximated based on specific experimental setups [43]. Applying convection coefficient values from directed energy deposition simulations to the molten metal jetting process compromises modeling accuracy and reliability due to differences in heat source parameters, inert gas flow rates, print platform movement speeds, and material deposition rates between the two processes.

Forced convection significantly influences heat transfer in various processes, ranging from jet impingement cooling to laser cladding and additive manufacturing. Prior studies have extensively documented how forced inert gas, and impinging jets enhance heat transfer by promoting turbulence, altering boundary layer characteristics, and amplifying local heat transfer coefficients. O'Donovan and Murray [44], [45] detailed the importance of turbulence-induced vortices in impinging jet systems, highlighting their ability to create localized peaks in heat transfer efficiency. In manufacturing contexts, such as laser cladding and directed energy deposition, researchers have emphasized the necessity of incorporating forced convection effects into thermal simulations. Gouge et al. [46] and Heigel et al. [42] demonstrated that forced convection, caused by shielding gases or powder delivery systems, substantially impacts thermal gradients, residual stresses, and the overall thermal history of the processed material. Given the established roles of forced convection in related manufacturing processes, incorporating accurate convection

coefficients into molten metal jetting simulations is essential for reliable thermal modeling. These studies emphasize that accurate modeling of forced convection is critical for predicting microstructural changes and optimizing manufacturing parameters.

Despite this wealth of knowledge, no study has yet quantified the convection coefficient for the molten metal jetting process, a critical parameter for understanding heat transfer within the process. In molten metal jetting, printing occurs in an open environment, unlike powder bed processes where the additive part is surrounded by metal powder. Since the surface of the part being built on the moving print bed is exposed to air, the convective heat transfer mechanism cannot be neglected. In addition, the local argon shielding gas flow near the droplet formation zone and rapid movements of the print platform during fabrication may significantly affect convective heat losses. Through experimental investigation and thermal modeling, this study aims to quantify the convection coefficient for the molten metal jetting process and provide valuable insights into the heat transfer dynamics of this metal additive manufacturing technique. Accurate thermal simulations are crucial because they enable us to understand the complex relationships between process parameters, the resulting microstructure, and the final properties of the manufactured parts [17]. These simulations offer insights into optimizing process parameters, improving part quality, and extending the applications of molten metal jetting technology. This study contributes to the field of metal additive manufacturing by facilitating the development of more accurate and efficient process models, optimizing process parameters, and advancing the capabilities and applications of molten metal jetting. Accurately quantifying the convection coefficient is a key step toward developing dependable thermal models for molten metal jetting.

To the best of our knowledge, this is the first study to quantitatively determine convective heat transfer coefficients in molten metal jetting by combining thermocouple-based experiments with a modified fin theory analysis and CFD modeling. The modified fin theory formulation accounts for the non-uniform ambient temperatures in MMJ. This novel theoretical adaptation, combined with experimental validation and CFD, allows one to accurately quantify convection in scenarios that traditional fin models would

misrepresent. This integrated approach fills a critical gap in thermal modeling for molten metal jetting by providing experimentally validated convection coefficients under realistic printing conditions.

To address this research gap, the study is structured around three core objectives:

1. To better understand how the temperature along a part's height varies under the four convection conditions: (i) natural convection, (ii) forced argon flow combined with close proximity to the high-temperature printhead, (iii) print platform movements, and (iv) the combined effects of forced argon flow, printhead proximity, and the rapid print platform movements.
2. To predict the thermal gradient along the part's height under the four distinct convection conditions mentioned above, using fin theory-based theoretical analysis.
3. To create a computational model to predict the thermal profiles along the part's height.

In addressing these objectives, this study aims to establish a rigorous and validated understanding of the convection coefficient in molten metal jetting. The results are expected to advance macro-scale thermal simulations in molten metal jetting by improving predictive accuracy, deepening the understanding of the process–structure–property relationship, and supporting the development of higher-quality metal components for broader industrial applications.

In summary, this work introduces a novel, combined experimental–theoretical–computational framework to determine convection coefficients in molten metal jetting, enabling the first experimentally grounded convection modeling for this process. By adapting classical fin theory for the unique non-uniform ambient conditions of MMJ and

validating it with both CFD and experimental measurements, this study provides an innovative solution to a long-standing thermal modeling challenge.

The **Methods** section presents the theoretical development of the convection model, describes computational modeling for the four different convection conditions, and outlines the experimental setup and temperature measurement procedures. The **Results and Discussion** section presents findings on convection coefficient values and the effects of process setup factors – such as print platform motion, argon flow, printhead proximity – on heat transfer, based on detailed theoretical analysis, computational modeling, and experimental measurements. The **Conclusions** section summarizes the key insights gained from the study, emphasizing the contributions to the field of metal additive manufacturing, and proposes potential areas for future research.

2. Methods

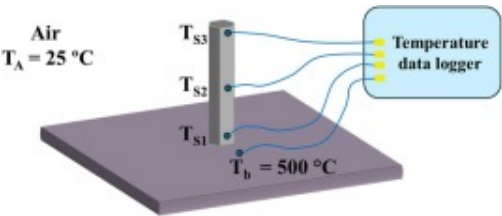
2.1. Experimental setup

A 13 cm tall hexagonal aluminum (Al 4008) part was printed using a Xerox ElemX metal 3D printer using the process illustrated in Fig. 1. The print parameters used during fabrication are listed in Table 1. Namely, the printhead's molten aluminum reservoir was maintained at 850°C, the heated build platform was held at 500°C, and the inert argon shielding gas flow rate was 7L/min. After the part was printed and while it remained on the heated platform, holes measuring 2.5 mm in diameter and 2 mm in depth were drilled on the side to accommodate thermocouples at three designated recording positions. The thermocouples were then secured using a high-performance silver paste. All three measurement points on the printed sample (S1, S2, and S3) were located on the side surface of the part, as shown in Fig. 3. Point S1 was located 10 mm above the print platform. Point S2 was at the sample midpoint (i.e., 65 mm from the print platform). Point S3 was 10 mm below the top surface of the part. A fourth thermocouple was positioned on the print platform to record the print bed temperature, T_b . The printed part was then subjected to four different convective cooling conditions. Real-time temperature data T_{S1} ,

T_{S2} , T_{S3} , and T_b , were recorded using K-type thermocouples during the cooling process. The ambient air temperatures near the side and top surfaces were also measured, as these values were later used as boundary conditions in the theoretical and computational models.

Table 1. Printing parameters for molten metal jetting process.

Print Parameter	Value
Molten aluminum reservoir temperature	850 °C
Print platform temperature	500 °C
Print platform speed	0.4m/s
Argon flow rate	7L/min
Droplet frequency	400Hz
Layer thickness	0.24mm

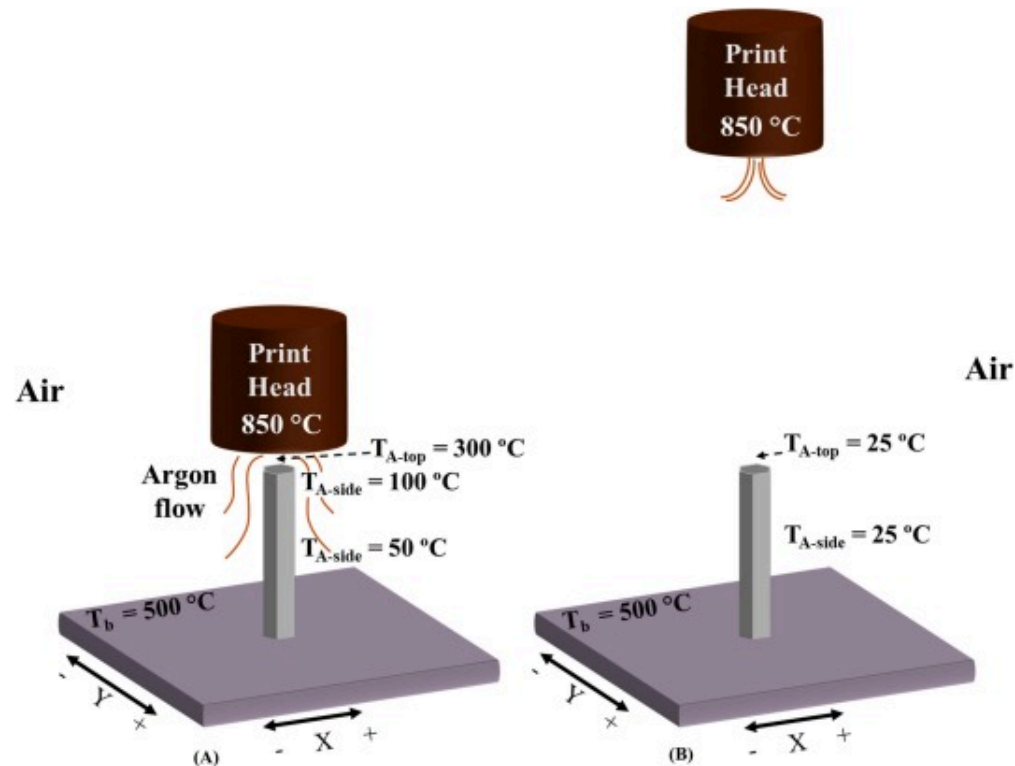


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Fig. 3. Temperature recording setup depicting locations of thermocouples T_{S1} , T_{S2} , and T_{S3} on the printed part, T_b is the print platform temperature, and T_A is the ambient temperature.

The following four convection conditions were evaluated to investigate the effects of natural convection, forced argon flow, printhead proximity, and platform motion, allowing their individual and combined contributions to convective heat transfer to be quantified:

1. *Natural convection*: The part was allowed to cool in air, with the surrounding temperature being 24°C. The print platform temperature was maintained at 500°C. The raised printhead position (20cm away from the top surface of the part) shown in Fig. 4(B) was used. Due to the large distance between the part and the printhead, argon flow from the print head has negligible effect on part cooling. In the following text, this convection condition is referred to as 'stationary platform and raised printhead (no argon)' condition.



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Fig. 4. Printing position (A) of printhead was 0.7cm away from the top surface of the part and non-printing position (B) was 20cm away from the top surface of the part. Orange path-lines represent heated argon flow. The indicated ambient temperatures are approximated.

2. *Mixed convection with argon flow and close printhead proximity:* The printhead was positioned 7 mm above the top surface of the stationary part to replicate the standoff distance during the printing process, as shown in [Fig. 4\(A\)](#). The argon gas flow rate was set to 7L/min. In the following text, this convection condition is referred to as the ‘stationary platform and lowered printhead (argon)’ condition.
3. *Enhanced convection due solely to print platform movements:* The print platform was subjected to rapid movements without nozzle heating or argon flow. The raised printhead position shown in [Fig. 4\(B\)](#) was used. Argon flow does not contribute to cooling due to the large distance between part and printhead. The print platform movement speed was 0.4m/s ([Table 1](#)). This speed corresponds to the standard build conditions for the ElemX machine used here and was used to replicate realistic operating conditions. In the following text, this convection condition is referred to as the ‘moving platform and raised printhead (no argon)’ condition.
4. *Combined effects of argon flow, printhead proximity, and print platform movements:* These three factors were combined to create the most dynamic convection condition, closely simulating the real-world molten metal jetting process. The lowered printhead position shown in [Fig. 4\(A\)](#) was used together with part movement and flowing argon. To replicate the platform motion during printing, the print platform was moved at a speed of 0.4m/s ([Table 1](#)), following alternating layer infill patterns. In the following text,

this convection condition is referred to as 'moving platform and lowered printhead (argon)' condition.

2.2. Theoretical analysis based on fin theory

We employ fin theory, a classical heat transfer model, to simulate the thermal behavior of parts fabricated through the molten metal jetting process, as it effectively accounts for extended surface cooling and heat dissipation mechanisms relevant to the process's open-environment conditions. While fin theory is well established, its application to molten metal jetting is novel because of the complex cooling dynamics introduced by open-environment heat loss mechanisms inherent to the process. By adapting fin theory to represent process-specific conditions, such as natural convection, forced argon flow, proximity to the high-temperature reservoir, and print platform movement, this study addresses a critical gap in understanding convective heat transfer in molten metal jetting. Natural convection facilitates heat loss due to air contact. Forced argon flow enhances localized cooling effects by increasing fluid turbulence. Proximity to the high-temperature molten metal reservoir can cause convective heating, while print platform movement promotes turbulence, resulting in a cooling effect. Collectively, these factors play a significant role in defining the heat transfer dynamics in molten metal jetting.

Fin theory is particularly suited for this analysis because it expresses complex heat transfer phenomena in the form of equations and the respective solutions, while remaining applicable to the open-environment setup in molten metal jetting. This approach accommodates the significant convective cooling observed in molten metal jetting by representing the heat transfer coefficient as a function of convection intensity and surface exposure. Surface area interactions directly influence cooling efficiency by determining the extent of heat exchange between the part and its surroundings, making them critical for optimizing thermal simulations.

2.2.1. Assumptions of fin theory and molten metal jetting

Assumptions were made in developing the theoretical model based on fin theory. These assumptions align closely with the conditions in the molten metal jetting process:

- One-dimensional heat conduction: The temperature gradient is evaluated only along the height of the part. Any temperature variation in the radial or transverse direction is considered negligible. Although one-dimensional heat conduction is assumed in the solid, convection takes place all around the solid surface on the external walls.
- Steady-state conditions: The printed part's temperature stabilizes over time, allowing for steady-state temperature distribution, which is fundamental to fin theory.
- Negligible radiation: Heat transfer is dominated by conduction and convection, with radiation losses assumed to be negligible at low to moderate operating temperatures involved (typically below 1000 °C) [10], [47]. This was confirmed here by comparing radiative and convective heat losses from the part's surface at mid-height (TS2) and near the top (TS3) using experimental surface temperature data ranging from 455 °C to 485 °C. A surface emissivity of 0.1 for the aluminum alloy was confirmed through thermal camera calibration. Applying the Stefan–Boltzmann law, the estimated net radiative heat flux ranged from 741.99 to 905.73 W/m². This is less than 10% of convective heat loss of 9844.1 to 10,264 W/m² found using a convection coefficient of 14 W/m²·°C. Radiation losses were therefore assumed to be minimal and were not considered in this study.
- Negligible contact resistance: Thermal resistance at the interface between the component and the base is assumed to be negligible.

These assumptions allow effective analysis of heat transfer dynamics in molten metal jetting parts, regardless of specific part geometry.

2.2.2. Theoretical model

In fin theory, the fundamental governing equation is the one-dimensional steady-state conduction equation with convective heat loss along the fin surface. For a fin with a uniform cross-sectional area [48], this equation is defined as follows:

$$\frac{d^2T}{dx^2} - \frac{hP}{kA_c}(T - T_\infty) = 0 \quad (1)$$

where T is the temperature at a position x mm away from the print platform along the height of the fin, T_∞ is the temperature of the surrounding (fluid) air, h is the convective heat transfer coefficient, P is the fin perimeter, k is the thermal conductivity of the part material, and A_c is cross-sectional area.

By defining the temperature difference as $\theta = T_b - T_\infty$, Eq. 1 can be written as:

$$\frac{d^2\theta}{dx^2} - m^2\theta = 0 \quad (2)$$

where $m^2 = hP/kA_c$. Eq. 2 is a linear second-order homogeneous differential equation. The general solution to Eq. 2 is:

$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx} \quad (3)$$

2.2.2.1. Constant ambient temperature assumption

At the base of the fin, $x = 0$, i.e., $\theta_b = \theta(x = 0) = T_b - T_\infty$. Applying Eq. 3 at the base of the fin yields

$$\theta_b = C_1 + C_2 \quad (4)$$

At the top of the fin, $\theta_L = \theta(x = L) = T_L - T_\infty$. Applying an energy balance, i.e., the rate of convective heat loss from the fin tip to the surrounding fluid equals the rate of conduction from the base to the fin-tip through the fin, yields:

$$hA_c[T_L - T_\infty] = -kA_c \left. \frac{dT}{dx} \right|_{x=L}$$

or

$$h \theta_L = -k \left. \frac{d\theta}{dx} \right|_{x=L} \quad (5)$$

Substituting Eq. 3 into Eq. 5,

$$h (C_1 e^{mL} + C_2 e^{-mL}) = -km (C_1 e^{mL} - C_2 e^{-mL}) \quad (6)$$

Solving for the constants C_1 and C_2 , the temperature distribution is derived using Eq. 3, Eq. 4, and Eq. 6 as follows:

$$\frac{\theta(x)}{\theta_b} = \frac{\cosh(m(L-x)) + \left(\frac{h}{mk}\right) \sinh(m(L-x))}{\cosh(mL) + \left(\frac{h}{mk}\right) \sinh(mL)} \quad (7)$$

where $\theta(x) = T_x - T_\infty$ is the temperature difference at position x along the part from print platform, $\theta_b = T_b - T_\infty$ is the temperature difference between the print platform and the surrounding environment, L is the length (height) of the printed part, and $m^2 \equiv hP/kA_c$. This theoretical analysis is based on Eq. 7, as it primarily models conduction within the fin while accounting for convective heat loss along the fin's surface.

2.2.2.2. Correction for non-uniform ambient temperature

Unlike conventional fin applications where ambient air is uniform, molten metal jetting involves strongly non-uniform ambient temperatures (cooler side surfaces and a much hotter top surface near the printhead). To our knowledge, this is the first adaptation of fin theory to account for such boundary conditions in additive manufacturing. The derivation in Section 2.2.2.1 assumes that the fin sides and the fin tip are exposed to the same ambient temperature. However, during the molten metal jetting process (Fig. 4 A), the side and top surfaces are exposed to different local ambient temperatures. In such a case, the governing equation (Eq. 2) remains unchanged, but the fin-tip boundary condition must be corrected.

The modified energy balance at the fin top is expressed as:

$$hA_c[T_L - T_{\infty, \text{top}}] = -kA_c \left. \frac{dT}{dx} \right|_{x=L} \quad (8)$$

where $T_{\infty, \text{top}}$ is the local ambient temperature at the fin top, which may differ from the side ambient, $T_{\infty, \text{side}}$. Substituting $\theta_L = T_L - T_{\infty, \text{side}}$ in Eq. 8, gives:

$$h[\theta_L - (T_{\infty, \text{top}} - T_{\infty, \text{side}})] = -k \left. \frac{d\theta}{dx} \right|_{x=L} \quad (9)$$

Eq. 9 is a mixed, or Robin-type [48], boundary condition at the fin top, which generalizes the standard convective-top formulation by allowing the local ambient temperature to differ from the side ambient temperature.

Substituting Eq. 3 into Eq. 9,

$$h(C_1 e^{mL} + C_2 e^{-mL} - (T_{\infty, \text{top}} - T_{\infty, \text{side}})) = -km(C_1 e^{mL} - C_2 e^{-mL}) \quad (10)$$

Solving for C_1 and C_2 , the temperature distribution can be shown as follows:

$$\frac{\theta(x)}{\theta_b} = \frac{\frac{h}{km} \sinh[m(L-x)] + \cosh[m(L-x)]}{\frac{h}{km} \sinh(mL) + \cosh(mL)} + \frac{T_{\infty, \text{tip}} - T_{\infty, \text{side}}}{\theta_b} \bullet \frac{\sinh(mx)}{\frac{h}{km} \sinh(mL) + \cosh(mL)} \quad (11)$$

Eq. 11 introduces a mixed boundary condition that accounts for a higher top surface ambient temperature, effectively reducing the apparent temperature gradient along the fin. A detailed derivation for Eq. 11 is available in Appendix A. The correction term in Eq. 11 accounts for the higher ambient temperature at the part's top surface, effectively reducing the apparent temperature gradient along the fin compared to the uniform-ambient case. In simple terms, because the air near the top of the part is hotter (especially when the printhead is nearby), the top of the part loses heat less rapidly, and the temperature drop from base to tip is not as steep as it would be under a uniform ambient temperature.

2.2.3. Boundary conditions for theoretical analysis

The following boundary conditions were employed:

- Base temperature T_b : The temperature of the print platform was maintained at a constant 500°C using external heating.
- Ambient temperature T_∞ : The surrounding environment's temperature varies depending on whether natural or forced convection is present. In this model, the printhead's position determines the type of convection condition. Natural convection occurs when the printhead is in a raised position far from the part. Under this condition, with raised printhead without argon flow, the ambient temperatures for the side (T_{A-side}) and top (T_{A-top}) of the part were both measured to be $24 \pm 1^\circ\text{C}$. On the other hand, forced convection is used under conditions when the printhead is lowered and is close to the part. In these cases, the part's ambient temperature rises due to the proximity of the high-temperature molten metal reservoir. Under the forced convection condition (lowered printhead with argon flow), ambient temperatures at the side (T_{A-side}) were recorded to be $49 \pm 2^\circ\text{C}$ at both the S1 and S2 locations, and $98 \pm 2^\circ\text{C}$ at the S3 location. The ambient temperature (T_{A-top}) for the top surface was measured to be $303 \pm 5^\circ\text{C}$.
- Convective heat transfer coefficient h : This coefficient is often derived in literature from empirical data or theoretical correlations (e.g., Nusselt numbers) for specific convection conditions. However, in this study, it was treated as a variable, and temperature distributions were calculated across a range of convection coefficients values.

2.2.4. Theoretical analysis

Fin theory based theoretical analysis was implemented in MATLAB, which provided an explicit solution framework. Prediction of temperature gradients along the part height was based on the steady-state heat conduction equation for one-dimensional heat

transfer (Eq. 1). Eq. 11 was solved across a range of convection coefficient values (6 to $12\text{W/m}^2\cdot^\circ\text{C}$). Three different ambient temperatures of $24\pm 1^\circ\text{C}$, $49\pm 2^\circ\text{C}$, and $98\pm 2^\circ\text{C}$ were considered. These ambient temperatures represent the surrounding air conditions, which vary based on the standoff distance between the heated molten metal jetting head and the top surface of the part, as well as the location of the temperature measurements. When the standoff distance between the jetting nozzle and top of the work piece is 20cm, the surrounding air temperature is 24°C on both sides and above the top surface of the part. When the standoff distance is 7mm from the top surface, proximity to the high-temperature printhead and the forced argon flow through the nozzle raise the surrounding air temperature to 49°C on the sides and 98°C above the top surface of the part. The calculation framework allows the theoretical analysis results to capture how different convection coefficients impact the cooling rate along the part under three distinct ambient temperature conditions. Temperature profiles were computed at 0.5mm intervals along the printed part to ensure sufficient data points for accurately capturing temperature gradients. These profiles reveal both the temperature gradient and the temperature differential between the base and top of the part under each boundary condition, providing a detailed view of the cooling behavior.

2.3. Computational modeling

A steady-state heat transfer computational model was set up in ANSYS Fluent to accurately represent the thermal behavior of the molten metal jetting process, including detailed geometry creation, mesh generation, material properties, and boundary conditions. The geometry of a 13cm tall hexagonal part (12.54mm side length) on the heated print platform was modeled in ANSYS DesignModeler. Mesh quality metrics were optimized, with a skewness of 0.3 and orthogonality exceeding 0.7, indicating a well-structured mesh. Skewness measures the deviation of the mesh elements from their ideal shape, while orthogonality assesses how closely element faces align with expected directions. Maintaining these values ensures accurate results by reducing computational errors and enhancing solution stability. To confirm mesh independence, a mesh convergence study was conducted using five progressively refined meshes, as

summarized in Table 2. The temperature values at three distinct locations (Fig. 3) on the side surface of the part — T_{s1} (10mm from the top surface), T_{s2} (65mm above the print platform), and T_{s3} (10mm from the print platform) — varied less than 1 °C across the tested meshes, demonstrating insensitivity to further mesh refinement. Based on these findings, a mesh with 71,109 elements was selected for subsequent simulations. This mesh density offered an optimal balance between computational efficiency and solution accuracy, ensuring robust and reliable results aligned with the study's objectives. The model was defined as a pressure-based steady state model to simulate energy (i.e., heat transfer). The governing equation for the model is:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q \quad (12)$$

where, ρ =density (kg/m³), c_p =specific heat capacity (J/kg·K), T =temperature (K), k =thermal conductivity (W/m·K), Q =volumetric heat source (W/m³), t =time (s). For steady-state simulation without heat generation, $\nabla \cdot (k \nabla T) = 0$.

Table 2. Mesh density and the resulting temperature values at three distinct locations across the length of the part.

Mesh Density (Elements)	T_{s1} (°C)	T_{s2} (°C)	T_{s3} (°C)
14,889	464.45	472.69	491.42
21,765	464.47	472.71	491.44
71,109	464.51	472.74	491.50
273,020	464.73	472.95	491.67
843,110	464.83	473.07	491.82

The ANSYS Fluent simulations were run to convergence, defined as reaching scaled residuals below 1×10^{-8} for the energy, momentum, and continuity equations. Under these criteria, the solutions achieved balance of energy with negligible residual

fluctuations, ensuring the results are fully converged. To maintain consistency with the theoretical model assumptions, radiative heat transfer was neglected in the CFD simulations. Convective and conductive heat transfer thus accounted for all cooling pathways in the model.

Boundary conditions were applied to define the fluid temperature at the top and sides of the domain, as recorded during experiments (Fig. 4). The print platform temperature was set to 500°C. Key material properties for Al 4008 are summarized in Table 3 [49], [50], [51].

Table 3. Temperature dependent values of the material properties of Al 4008.

Property	Temperature Dependent Value
Thermal Conductivity	$k_{solid} = 149.7 + 0.0809T - 0.0001T^2$ W/m·°C
Solidus Temperature	555°C
Liquidus Temperature	615°C

Initially, a full factorial parametric study was conducted using the parameter levels listed in Table 4. Subsequently, refined convection coefficient values were selected based on experimental trends to generate temperature profiles with improved agreement between computational and measured data.

Table 4. Computational modeling variables and the associated levels.

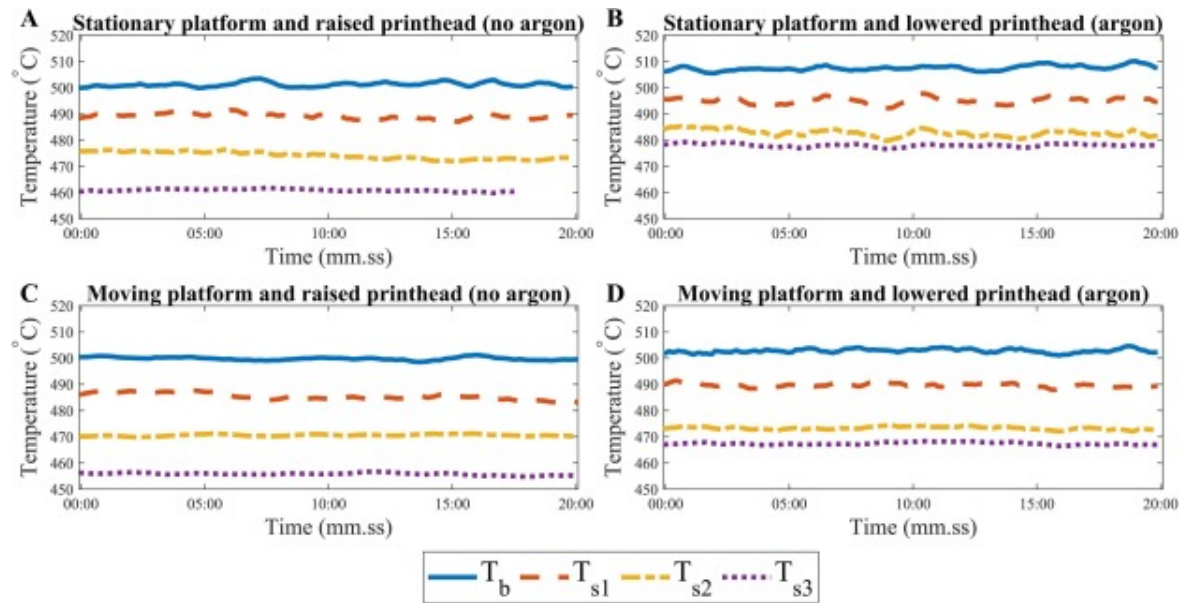
Variable	Values
Convection coefficient on the top surface of the part, h_{top}	8, 10, 12 W/m ² ·°C
Convection coefficient on the sides of the part, h_{side}	8, 10, 12 W/m ² ·°C
Surrounding fluid temperature on the side, T_{A-side}	24, 49, 98°C

Variable	Values
Surrounding fluid temperature on the top, T_{A-top}	24, 303°C

3. Results and discussion

3.1. Experimental results

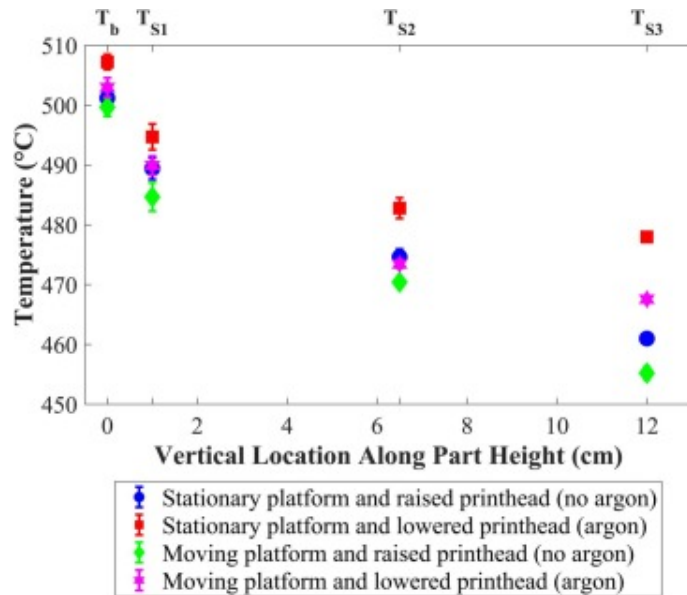
During the molten metal jetting process, the part experiences rapid print platform movement, a constant temperature at the heated base, argon flow over the top surface of the part being printed, and proximity to the high-temperature printhead due to a small standoff distance. These combined process factors create a complex heat transfer environment dominated by both conduction and convection. The resulting temperature gradients measured during experimental print trials showed a decrease in temperature along the part's height, with temperature decreasing as a function of distance from the heated print platform. To isolate the effects of these process factors on convective heat transfer, four distinct scenarios were analyzed, as shown in [Fig. 5](#) and [Fig. 6](#). [Fig. 5](#) displays time-dependent temperature stability at four specific locations (T_{S1} , T_{S2} , T_{S3} , and T_b) on the part surface over a 20-min period, indicating the steady-state conditions. Across three repeated trials, the standard deviation of measured temperatures at each thermocouple location was $\leq 4.72^\circ\text{C}$, consistent with the manufacturer's $\pm 2.22^\circ\text{C}$ calibration accuracy for K-type sensors. These uncertainties were later propagated into the best-fit convection coefficients reported in [3.2 Theoretical analysis based on fin theory](#), [3.3 Computational modeling](#). The baseline temperature at T_b consistently remained the highest, with temperatures progressively decreasing as a function of distance from the build plate, reflecting a heat dissipation pattern typical of the process. [Fig. 6](#) complements this temperature gradient data with average temperature values and error bars, providing comparative insights into spatial heat distribution.



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Fig. 5. Experimental temperature measurements at the four locations under different convection conditions: A. Natural convection; b. Forced convection with argon flow; c. Enhanced convection due to print platform movements; d. Combined effects of argon flow and print bed movements.



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Fig. 6. Average temperature for each experimental condition across measurement locations and convection conditions.

The first condition of natural convection was referred to as the stationary platform and raised printhead (no argon) case (Fig. 5(A)), where the part was allowed to cool under ambient air conditions, without any rapid print platform movements, argon flow, or close proximity to the high-temperature printhead. This condition resulted in a temperature gradient that decreased along the part's height as a function of distance from the print platform (Fig. 6, blue circles).

In the second condition (Fig. 5(B)), i.e., the stationary platform and lowered printhead (argon) case, the printhead was positioned at the standard process standoff distance of 7mm. Here, the combination of argon gas flow and printhead proximity increased local ambient temperatures near the top, thereby reducing the effective temperature gradient and producing a net heating effect. As a result, the recorded temperatures at all three

locations (T_{S1} , T_{S2} , and T_{S3}) were the highest among the four different convection conditions (Fig. 6, red squares), confirming the heating effect of argon flow from the closely positioned printhead.

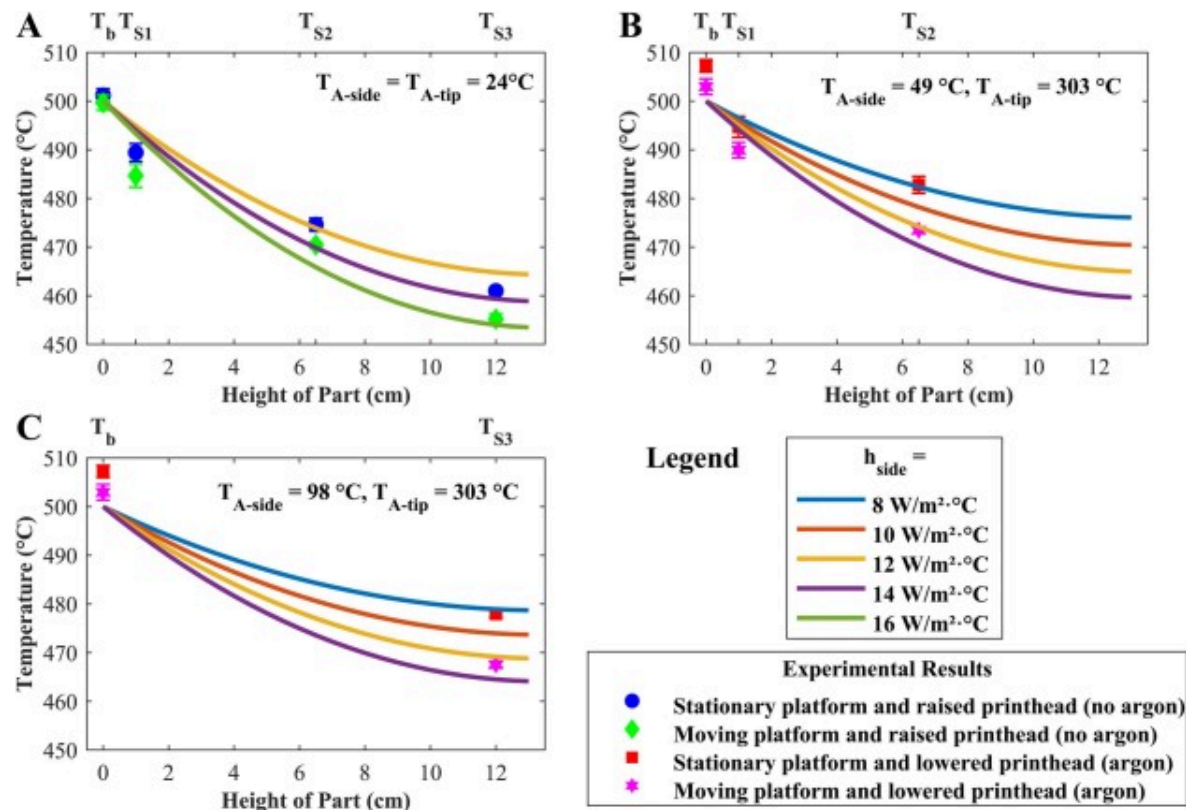
The third condition was the moving platform and raised printhead (no argon) case (Fig. 5(C)) where the standoff distance between the printhead and the part was 20cm. In this scenario, the heating effects of the argon flow and high-temperature printhead were eliminated. Instead, the rapid platform movement enhanced convective cooling by disrupting the boundary layer, allowing heat to escape more efficiently from the part's surface to the surrounding environment. As a result, the lowest temperature values among the four testing conditions were observed at all three locations (Fig. 6, green diamonds).

While platform movement alone (Fig. 5(C)) showed a cooling effect, and proximity to the printhead with forced argon flow showed a heating effect (Fig. 5(B)), combining platform movement with argon flow and printhead proximity (Fig. 5(D)) demonstrated a synergistic effect on heat dissipation. This fourth condition of moving platform and lowered printhead (argon) (Fig. 6, magenta star) produced higher temperatures at all three measurement points compared to the third condition (Fig. 6, green diamond), where only platform movement was involved. However, compared to the second condition with argon flow and printhead proximity without platform movement (Fig. 6, red square), this configuration showed lower temperatures at all three measurement points.

During the printing process, the argon flow and printhead proximity predominantly heated location S3 (near the top surface), whereas the cooling effects of platform movement are more pronounced at S1 and S2 (Fig. 6). The varied effects along the part's height highlight the complex interplay between thermal and flow dynamics in the molten metal jetting process. These experimental observations provide the baseline trends against the theoretical and computational models in subsequent sections.

3.2. Theoretical analysis based on fin theory

The fin theory-based model (Eq. 11) was implemented to predict the temperature distribution along the height of the printed part under four distinct convection conditions. Fig. 7 shows these predictions compared with the experimental measurements at thermocouple locations S1, S2, and S3. The theoretical predictions were generated for a range of convection coefficient values (h_{side}) at three representative side ambient temperature (T_{A-side}), with the top surface ambient temperature (T_{A-top}) set separately to account for printhead proximity. In Fig. 7(A), both T_{A-side} and T_{A-top} were 24°C, representing the stationary platform and raised printhead (no argon) case as well as the moving platform and raised printhead (no argon) case. Fig. 7(B) corresponds to the S2 location with T_{A-side} of 49°C and T_{A-top} of 303°C for the case with a stationary platform and lowered printhead (argon) (red squares), and the case with a moving platform and lowered printhead (argon) (magenta stars). Fig. 7(C) illustrates the condition for the S3 location where T_{A-side} is 98°C while T_{A-top} is 303°C. This represents the stationary platform and lowered printhead (argon) case (red squares), and the moving platform and lowered printhead (argon) case (magenta stars). Comparing theoretical predictions with experimental values allows estimation of effective convection coefficients under different process configurations.



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Fig. 7. Temperature profiles along the height of the part predicted using Fin Theory for different h_{side} values at ambient temperatures of (A) 24°C, (B) 49°C, and (C) 98°C. Experimental measurements at positions T_{S2} and T_{S3} are compared with the predictions under their respective ambient temperatures as boundary conditions.

In the absence of argon near the part (Fig. 7(A)), the best-fit convection coefficients (h_{side}) at S2 were 12W/m²·°C for the stationary platform and 14W/m²·°C for the moving platform. At S3, the values increased slightly to 13.5W/m²·°C and 15.5W/m²·°C. This reflects stronger cooling near the top surface compared to mid-height locations on the part surface. When the printhead was lowered with argon flow (Fig. 7(B)), the

corresponding values decreased; at S2 the best fits were $8\text{W/m}^2\cdot^\circ\text{C}$ for the stationary platform and $12\text{W/m}^2\cdot^\circ\text{C}$ for the moving platform. At S3, the results followed a similar trend (Fig. 7(C)), with a slight increase in h_{side} values. These results show that the model consistently predicted higher coefficients under platform motion (enhanced cooling) and lower coefficients under conditions dominated by argon flow and printhead proximity (localized heating). This demonstrates that even a simplified fin-theory model, when corrected for non-uniform ambient temperatures, can reproduce the relative influence of platform motion, argon flow, and printhead proximity on convective trends. The most realistic printing scenario, which is the moving platform with argon flow and printhead proximity, was characterized by a convection coefficient of $12.5\text{W/m}^2\cdot^\circ\text{C}$. This value yielded a good fit to the experimental temperature values at locations S2 (Fig. 7(B)) and S3 (Fig. 7(C)).

For the present geometry and fitted convection coefficients range of 8 to $14\text{W/m}^2\cdot^\circ\text{C}$, the fin parameter ranges from 0.405 to 0.536 , placing the part in a short-fin regime with modest but measurable axial temperature gradients. The corresponding Biot numbers range from 2.93×10^{-4} to 5.13×10^{-4} . These values are far below 0.1 , confirming that the one-dimensional conduction assumption used in the fin analysis is valid for this study [48]. Here, the fin parameter is calculated as mL , where L is the height of the part, and $m^2 \equiv hP/kA_c$ where h is the convective heat transfer coefficient, P is the fin perimeter, k is the thermal conductivity of the part material, and A_c is cross-sectional area. Biot numbers are calculated as $Bi = hL_c/k$ where $L_c = A_c/P$.

Discrepancies between the theoretical predictions and experimental values were most evident at location S1, the lowest thermocouple location near the platform. This could be due to the simplifying assumption of negligible thermal contact resistance at the part-platform interface. This limitation highlights the importance of accurately modeling thermal contact resistance in future work, especially for predictions near the platform as in the case of relatively short parts. The least-square errors between theoretical prediction curves for the displayed values in Fig. 7 and measured temperatures at locations S2 and S3 for each convection condition are summarized in Table 5.

Table 5. Least-square error between fin theory predictions and experimental measurements at T_{S2} and T_{S3} for four different convection conditions.

Convection Condition	Least-Square Error	
	T_{S2}	T_{S3}
Stationary platform and raised printhead (no argon)	0.513	4.684
Moving platform and raised printhead (no argon)	1.780	4.430
Stationary platform and lowered printhead (argon)	1.098	1.829
Moving platform and lowered printhead (argon)	2.613	4.656

3.3. Computational modeling

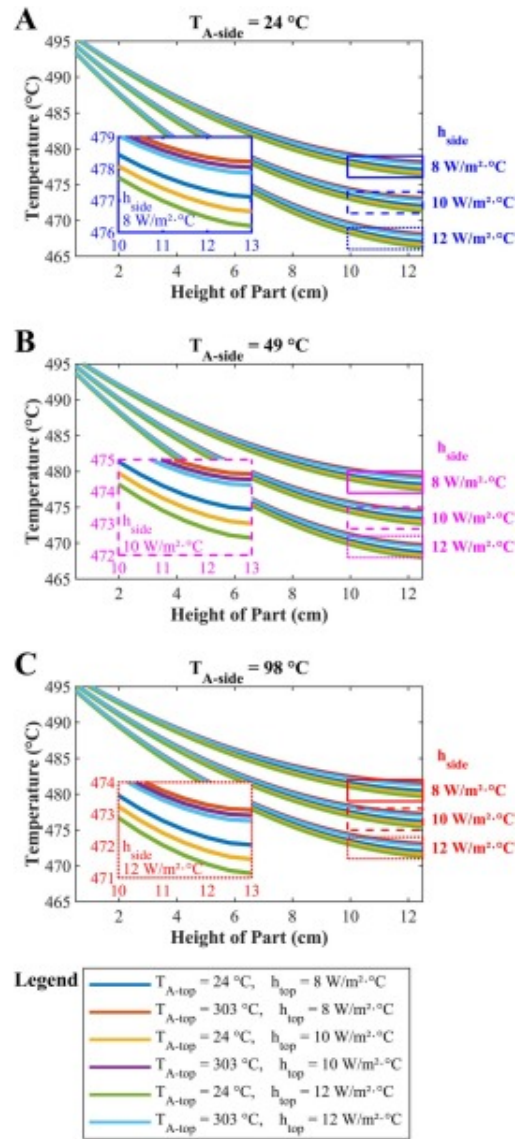
3.3.1. Computational model setup and boundary conditions

In the computational model developed using Ansys Fluent, the boundary conditions were defined using four parameters: the ambient temperature for the side walls of the part (T_{A-side}), the ambient temperature for the top surface (T_{A-top}), the convection coefficient for the side walls (h_{side}), and the convection coefficient for the top surface (h_{top}). The ambient temperatures (T_{A-side} and T_{A-top}) varied across the four convection conditions. For the stationary platform and raised printhead (no argon) case (i.e. blue circle and green diamond in Fig. 7), both T_{A-top} and T_{A-side} were 24°C. In the conditions which included forced argon flow and close proximity between the part's top surface and the hot printhead (red square and magenta star in Fig. 7), T_{A-top} was 303°C and T_{A-side} was either 49°C or 98°C, depending on the printhead's proximity and platform motion. These temperature values were applied as boundary conditions in the simulation to replicate the thermal environment of the printing process, where the effect of the argon jet and printhead proximity were studied. The computational model offered the ability to define ambient temperature and convection coefficient values for the top surface of the part,

allowing independent assignment of convection coefficients and ambient temperatures for the top and side surfaces. This provided a higher-fidelity representation of the molten metal jetting environment.

3.3.2. Parametric study results: thermal gradients and ambient effects

To investigate how boundary parameters affect temperature distributions, a full factorial study was conducted across the range of ambient temperatures and convection coefficients. [Fig. 8](#) shows the predicted thermal gradients from a full factorial parametric study using four parameters ([Table 4](#)): h_{side} , h_{top} , T_{A-side} , and T_{A-top} .



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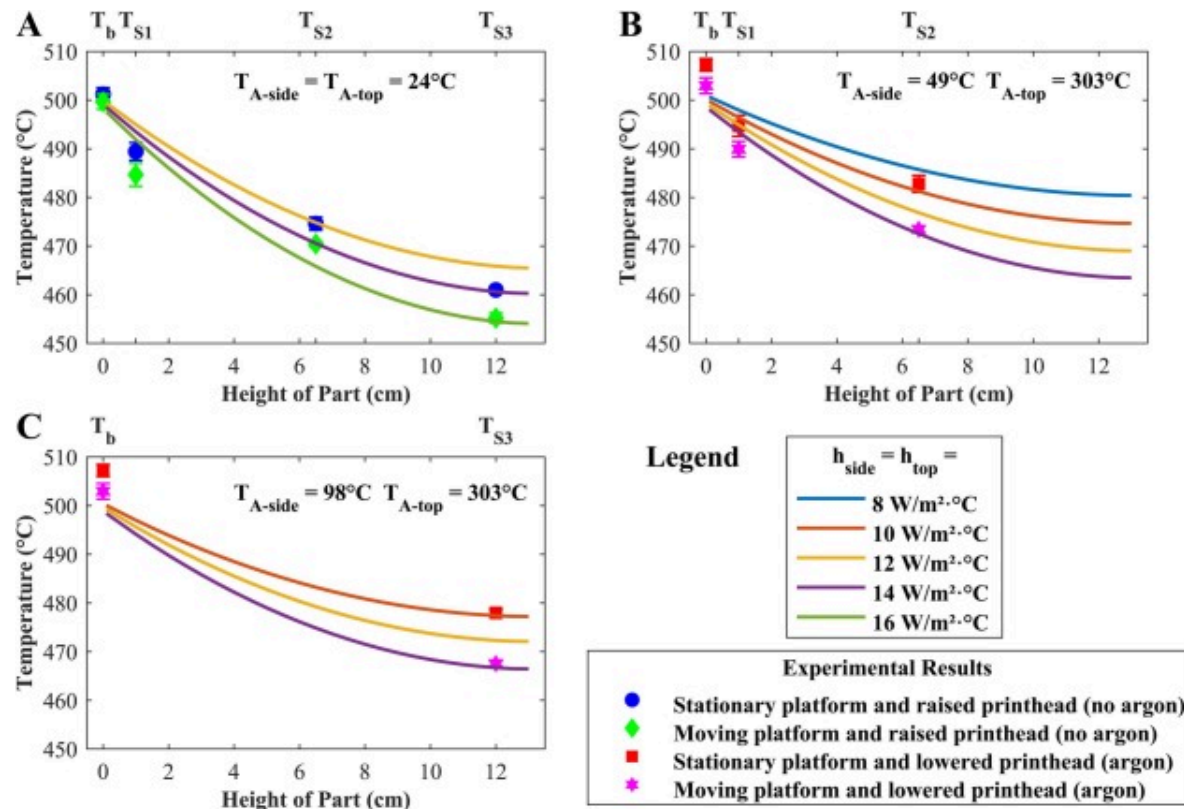
Fig. 8. Computational modeling results of temperature distribution along the height of a part for different convective heat transfer coefficients (h_{top}) and side temperatures (T_{A-side}). Results are shown for (A) $T_{A-side}=24\text{ }^{\circ}\text{C}$, (B) $T_{A-side}=49\text{ }^{\circ}\text{C}$, and (C) $T_{A-side}=98\text{ }^{\circ}\text{C}$ with T_{A-}

$T_{\text{top}}=24^{\circ}\text{C}$ and 303°C . Insets highlight temperature variations from 10cm to 13cm for finer details.

Across all three plots in Fig. 8, the temperature decreased as a function of increasing position along the part height, mirroring the trends observed in the experimental results (Section 3.1) and the theoretical analysis based on fin theory (Section 3.2). Higher convection coefficients resulted in steeper temperature gradients along the height of the part, signifying enhanced convective cooling. Overall temperature profiles were lower when $T_{A\text{-side}}=24^{\circ}\text{C}$ compared to $T_{A\text{-side}}=49^{\circ}\text{C}$ or $T_{A\text{-side}}=98^{\circ}\text{C}$ since the higher $T_{A\text{-side}}$ reduced the driving temperature difference between the part surface and surroundings. The inset plots in Fig. 8 further illustrate how increasing h_{top} enhanced convective cooling, while raising $T_{A\text{-top}}$ from 24°C to 303°C shifted the overall thermal profile to higher temperatures. For example, at $T_{A\text{-top}}=24^{\circ}\text{C}$, the temperature profile for $h_{\text{top}}=8\text{W/m}^2\cdot^{\circ}\text{C}$ (dark blue line) was higher than that for $h_{\text{top}}=12\text{W/m}^2\cdot^{\circ}\text{C}$ (green line) as expected, since higher h_{top} values enhance convective cooling at the top surface. Similarly, for $h_{\text{top}}=10\text{W/m}^2\cdot^{\circ}\text{C}$, the temperature profile at $T_{A\text{-top}}=24^{\circ}\text{C}$ (yellow line) shifted significantly lower than the profile at $T_{A\text{-top}}=303^{\circ}\text{C}$ (purple line). These trends were consistent across all three cases of $T_{A\text{-side}}$.

3.3.3. Validation against experimental data

To evaluate the accuracy of the computational approach, predicted temperature profiles were compared against experimental data at locations S2 and S3. The simulations were run with different convection coefficient values, and the values that produced the best agreement with the experimental data were identified. These selected convection coefficient values are shown in Fig. 9: along with the respective $T_{A\text{-side}}$ and $T_{A\text{-top}}$ values.



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Fig. 9. Comparison of computational temperature profiles with experimental measurements under different convection conditions. Symbols represent experimental data; solid lines correspond to computational predictions. Results are shown for (A) $T_{A-side} = T_{A-top} = 24^\circ\text{C}$, (B) $T_{A-side} = 49^\circ\text{C}$, $T_{A-top} = 303^\circ\text{C}$, and (C) $T_{A-side} = 98^\circ\text{C}$, $T_{A-top} = 303^\circ\text{C}$ for varying heat transfer coefficients: $8\text{ W/m}^2\cdot^\circ\text{C}$ to $16\text{ W/m}^2\cdot^\circ\text{C}$.

The results in Fig. 9: show that the convection coefficients at locations S2 and S3 varied across conditions, indicating localized differences in convective heat transfer along part's height. For the convective cases where the part was not exposed to argon flow and the high-temperature printhead (blue circle and green diamond in Fig. 9), the difference

between convection coefficients for locations S2 and S3 was about $2\text{ W/m}^2\cdot^\circ\text{C}$, consistent with stronger cooling near the top surface (S3) compared to the mid-height (S2). In contrast, the stationary platform and lowered printhead (argon) case (red square and magenta star in Fig. 9), showed a smaller difference of approximately $1\text{ W/m}^2\cdot^\circ\text{C}$ between convection coefficients for locations S2 and S3. This suggests that proximity to the printhead and exposure to forced argon flow create more uniform convective conditions along the part height, reducing the gradient in convection coefficient values.

The effect of print platform movement was also evident in the convection coefficient values. For the moving platform and raised printhead (no argon) case (green diamond in Fig. 9), the best fit convection coefficient value was $\sim 2\text{ W/m}^2\cdot^\circ\text{C}$ higher than the values identified for the stationary platform and raised printhead (no argon) case (blue circle in Fig. 9). This enhanced cooling was due to the print platform movement disrupting the surrounding fluid. Comparing the stationary platform and lowered printhead (argon) case (red square in Fig. 9) with the moving platform and lowered printhead (argon) case (magenta star), platform motion increased the convection coefficient by about $4\text{ W/m}^2\cdot^\circ\text{C}$ at both S2 and S3 locations. These results demonstrate that the cooling effect of platform motion persists even when superimposed on the heating influence of argon flow with printhead proximity.

The observed difference between convection coefficients at S2 and S3 in the raised printhead cases ($\Delta h \approx 2\text{ W/m}^2\cdot^\circ\text{C}$) is consistent with classical laminar natural convection and wake flows over vertical surfaces, where local Nusselt numbers increase with distance from the base [52]. Platform motion amplifies this trend, as boundary-layer disruption is more effective near the free upper edges of the part, increasing h at S3 relative to S2. In contrast, when the printhead is lowered and hot argon surrounds the top surface, the elevated local ambient temperature suppresses the effective ΔT at S3, counteracting the natural increase in h with height. This explains why fitted convection coefficients at S2 and S3 are nearly uniform in argon cases, consistent with the effect of mixed convection where forced heating reduces vertical variation.

For the moving platform and lowered printhead (argon) case (magenta star), the best fitted convection coefficient value was $14\text{W/m}^2\cdot^\circ\text{C}$. Overall, by identifying appropriate convection coefficient values, the computational modeling results reproduced the experimental temperature trends well, especially at locations S2 and S3. The predicted temperature values followed the same trends across all four conditions, with differences that can be explained by physical effects like localized cooling or heating. While T_{S1} showed larger deviations due to contact resistance between the part and the print bed, the model still provided a good match at the other two locations. The least-square errors for each convection condition are summarized in [Table 6](#).

Table 6. Least-square error between computational predictions and experimental measurements at T_{S2} and T_{S3} for four convection conditions.

Convection Condition	Least-Square Error	
	T_{S2}	T_{S3}
Stationary platform and raised printhead (no argon)	0.1702	1.1449
Moving platform and raised printhead (no argon)	0.0134	4.7786
Stationary platform and lowered printhead (argon)	6.15089	0.3249
Moving platform and lowered printhead (argon)	1.3409	0.6675

3.3.4. Comparison with theoretical analysis

Having established the best-fit convection coefficients for each case using the computational model, the results were then compared with the analytical predictions obtained from fin theory. [Fig. 7](#) presents theoretical profiles, while [Fig. 9](#) shows the computational results, both benchmarked against experimental measurements. For the stationary platform and raised printhead (no argon) and moving platform and raised printhead (no argon) cases (blue circle and green diamond), the convection coefficients

from the two approaches differed by $1 \text{ W/m}^2\cdot^\circ\text{C}$. For the conditions with argon present, the difference between the theoretical and computational predictions was slightly larger, on the order of 1 to $2 \text{ W/m}^2\cdot^\circ\text{C}$.

In both models, the lowest convection coefficients occurred when the platform was moving with the printhead raised away from the top surface, while the highest coefficients were obtained when the heated printhead was lowered close to the part. This consistent ranking reinforces the physical understanding that platform motion enhances convective cooling, whereas argon flow and printhead proximity elevate local ambient temperatures, effectively reducing the convection coefficient. The offset in absolute values can be attributed to the computational model's ability to define boundary conditions more accurately.

Although theoretical analysis and computational modeling differ in complexity and assumptions, both reliably reproduced the observed experimental thermal trends. Fin theory offers a quick and interpretable method for estimating effective convection coefficients, whereas the computational model provides high-fidelity simulations suitable for detailed process design and optimization. Together, the two approaches offer complementary advantages for understanding convective heat transfer in molten metal jetting.

The computational model successfully captured the effects of platform motion, argon flow, and printhead proximity on heat transfer within the molten metal jetting process. By enabling localized control of ambient conditions and convection coefficients, the model offered high-fidelity predictions that aligned well with experimental measurements, especially at the mid and top locations of the part. These results underscore the utility of computational modeling as a precise and adaptable tool for simulating convective heat transfer in additive manufacturing environments.

3.4. Flow regime analysis

The shielding argon jet and its induced downwash along the sidewall, the platform-induced cross-flow across the part diameter, and the buoyant rise along the vertical walls each operate in distinct flow regimes, with different impacts on heat transfer. Thermophysical properties of argon [53] and air [48] at the film temperature (defined as the average of the surface and bulk gas temperatures) were used to characterize these regimes (Appendix B.1).

For the condition with a raised printhead without platform motion, the Rayleigh number along the 13 cm wall is $Ra \approx 8 \times 10^6$ (Appendix B.2), well below the laminar-turbulent transition threshold for natural convection ($Ra \approx 10^9$ [52]), confirming that sidewall convection is laminar in this baseline case [54]. Physically, this corresponds to buoyant air rising upward along the sidewalls of the part, with a laminar boundary layer forming from the base and thickening from the base toward the top. When platform motion was introduced, the wake Reynolds number based on part diameter $D=25.4\text{mm}$ and raster speed $U_{platform} \approx 0.4\text{m}\cdot\text{s}^{-1}$ was $Re_D \approx 251$ (Appendix B.4). Since wake flows remain steady only up to $Re \approx 32$ [52], the raster motion produced unsteady vortex shedding (Kármán street [56]) and traditional wakes [55], disrupting the thermal boundary layers and increasing convective cooling. As a result of platform motion, convection coefficients increased by $2\text{--}4\text{W}/\text{m}^2\cdot^\circ\text{C}$ (Fig. 9(A)).

For the argon annular jet, the nozzle Reynolds number based on the hydraulic diameter $D_h=7\text{mm}$ and mean exit velocity $U_{jet} \approx 1.1\text{m}\cdot\text{s}^{-1}$, is $Re_{Dh} \approx 142$ (Appendix B.3). Since round jets remain laminar only up to $Re < 30$ [52], the argon jet becomes turbulent almost immediately downstream of the nozzle. Upon impingement on the part top surface, the argon flow turns downward and forms a wall-tangential downwash along the sidewall. The balance between forced downwash and buoyancy was evaluated using the mixed-convection parameter $M(z)$. The criterion yields $M(z) < 1$ over the entire 13 cm wall (Appendix B.5), confirming that forced downwash dominates buoyancy. This strong downwash suppresses the natural increase of convection coefficient with height, consistent with the observation that S2 (Fig. 9(B)) and S3 (Fig. 9(C)) values were nearly uniform in the lowered printhead cases. The turbulent jet did not enhance convective

cooling, instead it elevated the local ambient temperature around the top surface, reducing the effective driving ΔT and suppressing the increase of h with height.

The identified flow regimes aligned with interpretations of the fitted coefficients when expressed as Nusselt numbers (Appendix B.6). For the raised printhead and stationary platform condition, fitted coefficient values $12\text{W/m}^2\cdot^\circ\text{C}$ and $14\text{W/m}^2\cdot^\circ\text{C}$ at S2 and S3 corresponded to $Nu_L \approx 40$ and 47, values that were inverted using Churchill-Chu correlation [57] to the laminar buoyant regime numbers: $Ra_L \approx 3.42 \times 10^7$ and 6.58×10^7 . With platform motion, the coefficients increased to $12\text{W/m}^2\cdot^\circ\text{C}$ and $14\text{W/m}^2\cdot^\circ\text{C}$ for S2 and S3 locations, giving respective $Nu_L \approx 47$ and 53, which were inverted to $Re_D \approx 416$ and 544 from the Zukauskas correlation [58], confirming the unsteady laminar to early transitional shedding regime. Thus, buoyancy drove an upward laminar boundary layer while raster motion superimposed a lateral wake that modestly increased convective transport. Here, the platform motion increased Nu_L by 17% at S2 and 13% at S3.

For the actual printing condition with lowered printhead with platform motion, where platform motion and argon downwash occurred simultaneously, fitted coefficient of $14\text{W/m}^2\cdot^\circ\text{C}$ corresponded to $Nu_D \approx 9.4$. The effective $Re_D \approx 518$ was calculated from Nu_D value using Zukauskas correlation, confirming the transitional shedding regime identified independently from geometry-based estimates (Appendix B.4). Platform motion in this case produces only a $\sim 10\%$ increase midway up the part's height (S2) and essentially no change at the top (S3). Little to no increase in Nu_L values in cases of a lowered printhead confirmed that the strong wall-tangential downwash suppresses the natural increase of Nu_L with height. This was also independently observed by analyzing the mixed-convection parameter $M(z)$. Thus, the combined action of argon downwash and raster cross-flow during printing maintains the part in a transitional convective state, explaining both the modest enhancement with platform motion and the suppression of vertical variation compared to air-only cases.

4. Conclusions

The research presented here is the first-known study of convective heat transfer for an emerging additive manufacturing process known as molten metal jetting. The process has unique elements that complicate heat transfer coefficient estimation, such as a moving heated build platform that causes relative motion of the part through the ambient surrounding atmosphere, the proximity of a high temperature molten metal jetting assembly, and the localized downward flow of hot argon gas that prevents oxidation of the molten metal droplets during deposition. Each of these factors were considered in this research, leading to the following key findings:

- An adaptation of classic fin theory that incorporated multiple ambient temperature values based upon proximity of the part surface(s) to the heated molten metal print head was employed to obtain good agreement with experimentally validated ANSYS simulation results.
- Under the scenario where the surface of a part is moving during printing and is in close proximity to the heated printhead with flowing argon gas, convection heat transfer coefficients of 12.5 and 14.0W/m².°C were found for fin theory and experimentally validated ANSYS simulation results respectively.
- In the above scenario, the convection coefficient values were consistent regardless of vertical distance along the part surface from the print bed.
- For areas of a part being printed that are not in close proximity to the heated print head (i.e. when printing a very large part), the scenario where the platform is moving with a raised head and no argon flow is more appropriate. In this scenario, convection heat transfer coefficients of 15.5 and 16.0W/m².°C were found for fin theory and experimentally validated ANSYS simulation results respectively at the top surface of the part. Convection coefficients dropped to 14.0W/m².°C for surfaces below the top of the part. The slightly higher cooling under this scenario is due to absence of the heated argon flow around the print head.

Looking ahead, these coefficients will be of considerable value in process parameter optimization studies. Specifically, MMJ produces parts via jetting of discrete molten metal droplets which land, spread out, and solidify. Multi-physics simulations that incorporate heat transfer via conduction and the newly obtained convection coefficients will allow researchers to evaluate the effect of alternative drop placement strategies on microstructure, geometric accuracy, and porosity when building parts. For example, process parameter optimization studies involving controllable parameters such as droplet jetting frequency, droplet overlap distance, and droplet temperature can create scenarios with widely varying thermal history conditions that influence microstructure, dimensional accuracy, and pore formation in MMJ parts. Slower material cooling rates can result in coarse microstructures and in extreme cases, shape deviation from the original CAD geometry due to surface tension effects. In contrast, higher cooling rates can lead to a refined microstructure with improved strength. However, excessively high cooling rates can result in droplets solidifying before they have spread out and merged with previously deposited material, thus producing porous parts. The inclusion of convection heat transfer coefficients developed in this work will enable more accurate thermal simulations used in these process parameter optimization studies to balance outcomes related to microstructure, dimensional accuracy, and density/porosity.

CRediT authorship contribution statement

Khushbu Zope: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Isaac Perez-Raya:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Formal analysis. **Usama Abdullah Rifat:** Writing – review & editing, Methodology. **Paarth Mehta:** Writing – review & editing, Methodology. **Denis Cormier:** Writing – review & editing, Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Denis Cormier reports financial support was provided by Empire State Development Corp Division of Science Technology and Innovation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Analytical Correction Applied to Fin Theory

The general solution to a linear second-order homogenous differential form of the fundamental governing equation is given by Eq. 3, which is -

$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx} \quad (\text{A.13})$$

Derivative:

$$\frac{d\theta}{dx} = mC_1 e^{mx} - mC_2 e^{-mx} \quad (\text{A.14})$$

At $x=L$:

$$\theta_L = C_1 e^{mL} + C_2 e^{-mL} \quad (\text{A.15})$$

$$\left. \frac{d\theta}{dx} \right|_{x=L} = mC_1 e^{mL} - mC_2 e^{-mL} \quad (\text{A.16})$$

At base $x=0$:

$$\theta_b = C_1 + C_2 \quad (\text{A.17})$$

When the ambient temperatures are different for the fin sides and fin tip, the energy balance, i.e., the rate of convective heat loss from the fin tip to the surrounding fluid equals the rate of conduction from the base to the fin-tip through the fin, yields Eq. 8 from the main article. This equation is the modified boundary condition for the fin tip:

$$hA_c[T_L - T_{\infty,tip}] = -kA_c \left. \frac{dT}{dx} \right|_{x=L} \quad (A.18)$$

Substituting $\theta_L = T_L - T_{\infty,side}$, i.e., $T_L = \theta_L + T_{\infty,side}$ in Eq. A.6, we get Eq. 9:

$$h[\theta_L - (T_{\infty,tip} - T_{\infty,side})] = -k \left. \frac{d\theta}{dx} \right|_{x=L} \quad (A.19)$$

In Eq. A.7, substituting expressions for θ_L from Equation III and $\left. \frac{d\theta}{dx} \right|_{x=L}$ from Eq. A.4, we get Eq. 10:

$$h(C_1 e^{mL} + C_2 e^{-mL} - (T_{\infty,tip} - T_{\infty,side})) = -km(C_1 e^{mL} - C_2 e^{-mL}) \quad (A.20)$$

Rearranging the terms in Eqs. A.7, we get:

$$\begin{aligned} hC_1 e^{mL} + hC_2 e^{-mL} - h(T_{\infty,tip} - T_{\infty,side}) &= -kmC_1 e^{mL} + kmC_2 e^{-mL} \\ (h + km)C_1 e^{mL} + (h - km)C_2 e^{-mL} &= h(T_{\infty,tip} - T_{\infty,side}). \end{aligned} \quad (A.21)$$

From Eq. A.5, $C_2 = \theta_b - C_1$. Substituting this expression of C_2 in Eq. A.9,

$$\begin{aligned} (h + km)C_1 e^{mL} + (h - km)(\theta_b - C_1)e^{-mL} &= h(T_{\infty,tip} - T_{\infty,side}). \\ (h + km)C_1 e^{mL} + (h - km)\theta_b e^{-mL} - (h - km)C_1 e^{-mL} &= h(T_{\infty,tip} - T_{\infty,side}). \\ C_1 [(h + km)e^{mL} - (h - km)e^{-mL}] + (h - km)\theta_b e^{-mL} &= h(T_{\infty,tip} - T_{\infty,side}). \\ C_1 &= \frac{h(T_{\infty,tip} - T_{\infty,side}) - (h - km)\theta_b e^{-mL}}{(h + km)e^{mL} - (h - km)e^{-mL}}. \end{aligned} \quad (A.22)$$

The temperature distribution term $\frac{\theta(x)}{\theta_b}$ can be derived by normalizing equation by θ_b .

$$\frac{\theta(x)}{\theta_b} = \frac{1}{\theta_b} (C_1 e^{mx} + C_2 e^{-mx}).$$

Again substituting $C_2 = \theta_b - C_1$,

$$\begin{aligned}\frac{\theta(x)}{\theta_b} &= \frac{1}{\theta_b} (C_1 e^{mx} + (\theta_b - C_1) e^{-mx}) \\ \frac{\theta(x)}{\theta_b} &= \frac{C_1}{\theta_b} (e^{mx} - e^{-mx}) + e^{-mx}.\end{aligned}\quad (\text{A.23})$$

From Eq. A.10,

$$\frac{C_1}{\theta_b} = \frac{\frac{h}{\theta_b} (T_{\infty, \text{tip}} - T_{\infty, \text{side}}) - (h - km) e^{-mL}}{(h + km) e^{mL} - (h - km) e^{-mL}}. \quad (\text{A.24})$$

Substituting expression for $\frac{C_1}{\theta_b}$ in Eq. A.11,

$$\frac{\theta(x)}{\theta_b} = \left[\frac{\frac{h}{\theta_b} (T_{\infty, \text{tip}} - T_{\infty, \text{side}}) - (h - km) e^{-mL}}{(h + km) e^{mL} - (h - km) e^{-mL}} \right] (e^{mx} - e^{-mx}) + e^{-mx}. \quad (\text{A.25})$$

To simplify Eq. A.13, the following hyperbolic identities were used:

$$\begin{aligned}\cosh(z) &= \frac{e^z + e^{-z}}{2}, \quad \sinh(z) = \frac{e^z - e^{-z}}{2} \\ e^{mL} + e^{-mL} &= 2 \cosh(mL), \quad e^{mL} - e^{-mL} = 2 \sinh(mL)\end{aligned}$$

So, the denominator of Eq. A.13 can be rearranged as -

$$(h + km) e^{mL} - (h - km) e^{-mL} = 2h \sinh(mL) + 2km \cosh(mL). \quad (\text{A.26})$$

Substituting denominator by Eq. A.14 and the multiplier term by hyperbolic functions in Eq. A.13, we get

$$\frac{\theta(x)}{\theta_b} = \left[\frac{\frac{h}{\theta_b} (T_{\infty, \text{tip}} - T_{\infty, \text{side}}) - (h - km) e^{-mL}}{2(h \sinh(mL) + km \cosh(mL))} \right] \bullet 2 \sinh(mx) + e^{-mx}. \quad (\text{A.27})$$

$$\frac{\theta(x)}{\theta_b} = \frac{\frac{h}{\theta_b} (T_{\infty, \text{tip}} - T_{\infty, \text{side}}) \sinh(mx)}{h \sinh(mL) + km \cosh(mL)} - \frac{(h - km) e^{-mL} \sinh(mx)}{h \sinh(mL) + km \cosh(mL)} + e^{-mx}. \quad (\text{A.28})$$

In Eq. A.16, first term is the correction term, and last two terms are base solution terms.

Let us call base term $B(x)$,

$$B(x) = -\frac{(h-km)e^{-mL} \sinh(mx)}{h \sinh(mL) + km \cosh(mL)} + e^{-mx}.$$

$$B(x) = -\frac{(h-km)e^{-mL} \sinh(mx) + e^{-mx} h \sinh(mL) + e^{-mx} km \cosh(mL)}{h \sinh(mL) + km \cosh(mL)}.$$

$$B(x) = \frac{-he^{-mL} \sinh(mx) + km e^{-mL} \sinh(mx) + e^{-mx} h \sinh(mL) + e^{-mx} km \cosh(mL)}{h \sinh(mL) + km \cosh(mL)}. \quad (A.29)$$

Substituting exponentials (e^{-mx} and e^{-mL}) in terms of $\cosh$ and $\sinh$, Eq. A.17 reduces to:

$$B(x) = \frac{\frac{h}{km} \sinh[m(L-x)] + \cosh[m(L-x)]}{\frac{h}{km} \sinh(mL) + \cosh(mL)}. \quad (A.30)$$

where, $\sinh[m(L-x)] = \sinh(mL)\cosh(mx) - \cosh(mL)\sinh(mx)$ and

$$\cosh[m(L-x)] = \cosh(mL)\cosh(mx) - \sinh(mL)\sinh(mx)$$

Correction term from Eq. A.16, let us call it $A(x)$, can be rearranged as:

$$A(x) = \frac{\frac{h}{\theta_b} (T_{\infty, \text{tip}} - T_{\infty, \text{side}}) \sinh(mx)}{h \sinh(mL) + km \cosh(mL)}$$

$$A(x) = \frac{T_{\infty, \text{tip}} - T_{\infty, \text{side}}}{\theta_b} \bullet \frac{\sinh(mx)}{\frac{h}{km} \sinh(mL) + \cosh(mL)} \quad (A.31)$$

Substituting the rearranged base term (Eq. A.18) and rearranged correction term (Eq. A.19) into Eq. A.16, we get the corrected temperature distribution across the fin when the fin tip is subjected to a different ambient temperature than the side ambient temperature., as follows:

$$\frac{\theta(x)}{\theta_b} = \frac{\frac{h}{km} \sinh[m(L-x)] + \cosh[m(L-x)]}{\frac{h}{km} \sinh(mL) + \cosh(mL)} + \frac{T_{\infty, \text{tip}} - T_{\infty, \text{side}}}{\theta_b} \bullet \frac{\sinh(mx)}{\frac{h}{km} \sinh(mL) + \cosh(mL)}$$

Appendix B. Dimensionless numbers and mixed-convection calculations

B.1. Properties and geometry used

Argon	Temperature: $T_{\text{argon}} \approx 303^\circ\text{C}$ (573 K) Film temperature, $T_f = (T_{\text{surface}} + T_{\text{argon}})/2 = (474 + 303)/2 = 387^\circ\text{C}$ (660 K) Density: $\rho \approx 0.73 \text{ kg}\cdot\text{m}^{-3}$ Dynamic viscosity: $\mu \approx 4 \times 10^{-5} \text{ Pa}\cdot\text{s}$ Kinematic viscosity: $\nu = \mu/\rho \approx 5.5 \times 10^{-5} \text{ m}^2\cdot\text{s}^{-1}$ Thermal Conductivity: $k \approx 0.038 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ Specific heat: $c_p \approx 520 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$ Thermal Diffusivity: $\alpha = k/(\rho c_p) \approx 1.0 \times 10^{-4} \text{ m}^2\cdot\text{s}^{-1}$ Prandtl number: $Pr = \nu/\alpha \approx 0.55$ Thermal expansion coefficient: $\beta \approx 1/T_f \approx 1/660 \text{ K}^{-1} \approx 0.001508 \text{ K}^{-1}$
Air	Temperature: $T_{\text{air}} \approx 24^\circ\text{C}$ (298 K) Film temperature, $T_f = (T_{\text{surface}} + T_{\text{air}})/2 = (465 + 24)/2 = 245^\circ\text{C}$ (518 K) Density: $\rho \approx 0.679 \text{ kg}\cdot\text{m}^{-3}$ Dynamic viscosity: $\mu \approx 2.74 \times 10^{-5} \text{ Pa}\cdot\text{s}$ Kinematic viscosity: $\nu = \mu/\rho \approx 4.04 \times 10^{-5} \text{ m}^2\cdot\text{s}^{-1}$ Thermal Conductivity: $k \approx 0.039 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ Specific heat: $c_p \approx 1010 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$ Thermal Diffusivity: $\alpha = k/(\rho c_p) \approx 5.69 \times 10^{-4} \text{ m}^2\cdot\text{s}^{-1}$ Prandtl number: $Pr = \nu/\alpha \approx 0.71$ Thermal expansion coefficient: $\beta \approx 1/T_f \approx 1/518 \text{ K}^{-1} \approx 0.001923 \text{ K}^{-1}$
Part geometry:	Vertical height: $L = 0.13 \text{ m}$ Diameter for cross-flow: $D = 25.4 \text{ mm}$

Nozzle geometry: Annular nozzle: $D_o=13\text{ mm}$, $D_i=6\text{ mm}$
 Area: $A=(\pi/4)(D_o^2-D_i^2)=1.0446\times 10^{-4}\text{ m}^2$
 Perimeter: $P=\pi(D_o+D_i)$
 Hydraulic diameter $D_h=4A/P\approx 7.0\text{ mm}$
 Argon flow rate: $Q=7\text{ L}\cdot\text{min}^{-1}=1.1667\times 10^{-4}\text{ m}^3\cdot\text{s}^{-1}$
 Mean exit speed: $U_{jet}=Q/A\approx 1.117\text{ m}\cdot\text{s}^{-1}$

Process: Platform raster speed: $U_{platform}\approx 0.4\text{ m}\cdot\text{s}^{-1}$

B.2. Raleigh number for raised printhead cases

Air: $g=9.81\text{ m}\cdot\text{s}^{-2}$, $\beta\approx 0.001923\text{ K}^{-1}$, $\Delta T=445\text{ K}$, $L=0.13\text{ m}$, $\alpha=5.69\times 10^{-5}\text{ m}^2\cdot\text{s}^{-1}$,
 $v_{air}=4.04\times 10^{-5}\text{ m}^2\cdot\text{s}^{-1}$

$$Ra_L = \frac{g\beta\Delta TL^3}{\alpha v_{air}} = \frac{9.81\times 0.001923\times 445\times 0.13^3}{5.69\times 10^{-5}\times 4.04\times 10^{-5}} \approx 8.02 \times 10^6$$

B.3 Nozzle Reynold Number for Annular Argon Jet.

Argon: $U_{jet}\approx 1.117\text{ m}\cdot\text{s}^{-1}$, $D_h=7\text{ mm}$, $v_{argon}=5.5\times 10^{-5}\text{ m}^2\cdot\text{s}^{-1}$

$$Re_{D_h} = \frac{U_{jet} D_h}{v_{argon}} = \frac{1.117\times 0.007}{5.5\times 10^{-5}} \approx 142$$

B.4 Wake Reynold Number for Platform Raster Motion.

Air: $U_{platform}=0.4\text{ m}\cdot\text{s}^{-1}$, $D=0.0254\text{ m}$, $v_{air}=4.04\times 10^{-5}\text{ m}^2\cdot\text{s}^{-1}$

$$Re_{D,air} = \frac{U_{platform} D}{v_{air}} = \frac{0.4\times 0.0254}{4.04\times 10^{-5}} \approx 251$$

Argon: $U_{platform}=0.4\text{ m}\cdot\text{s}^{-1}$, $D=0.0254\text{ m}$, $v_{argon}=5.5\times 10^{-5}\text{ m}^2\cdot\text{s}^{-1}$

$$Re_{D,argon} = \frac{U_{platform} D}{v_{argon}} = \frac{0.4\times 0.0254}{5.5\times 10^{-5}} \approx 186$$

B.5 Mixed-convection balance parameter

Mixed- convection parameter $M(z)$ quantifies the competition between forced downwash from the argon flow and buoyancy along the vertical sidewall, where z is the distance on z axis starting from the nozzle on the printhead along the length of the part toward the print platform. $M(z)$ applies only to the wall-tangential flow created when the annular argon flow impinges on the top and turns over the edge. Platform raster motion is a cross-flow across the diameter and is characterized separately by the wake Reynolds number, not by the mixed connection parameter. $M(z) < 1$ forced downwash dominates, $M(z) \gtrsim 1$ buoyancy becomes competitive.

$$M(z) = \frac{Ra_z^{1/4}}{Re_z^{1/2} Pr^{1/3}}$$

z	$Re_z = \frac{U_{jet} z}{v_{argon}}$	$Ra_z = \frac{g\beta\Delta T z^3}{\alpha v_{argon}}$	$M(z)$
0.01 m	205	2.73×10^2	0.353
0.05 m	1025	3.41×10^4	0.527
0.1 m	2050	2.73×10^5	0.628
0.13 m	2665	5.996×10^5	0.671

$M(z) < 1$ over the full 13 cm, so forced jet downwash dominates along the sidewall for this speed.

B.3. Nusselt numbers from fitted convection coefficients

$$Nu = \frac{hL_c}{k}$$

Where h is the fitted convection coefficient, L_c the characteristic length for the mechanism, and k the thermal conductivity of the fluid at the film temperature.

Thermal conductivities at the film temperature:

Air ($T_f=518\text{K}$): $k \approx 0.039 \text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$.

Argon ($T_f=660\text{K}$): $k \approx 0.038 \text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$.

To connect the fitted convection coefficients to canonical flow-regime metrics, the Nusselt numbers were inverted to calculate (i) the effective Rayleigh numbers for the vertical sidewall cases using Churchill-Chu correlation and (ii) the wake Reynolds numbers for the platform-induced cross-flow using Zukauskas correlation. Film-temperature properties from Appendix B.1 were used for these calculations.

Churchill-Chu correlation for Raleigh number:

$$\text{Nu}_L = 0.68 + \frac{0.67 \text{Ra}_L^{1/4}}{\left[1 + (0.492 / \text{Pr})^{9/16}\right]^{4/9}}$$

From Churchill-Chu formula, Ra_L was calculated using $L=0.13\text{m}$ and $Pr_{air} \approx 0.71$.

Zukauskas correlation for cross-flow over cylindrical parts for Renolds number:

$$\text{Nu}_D = C \text{Re}_D^m \text{Pr}^n \left(\frac{\text{Pr}}{\text{Pr}_s}\right)^{0.25}$$

where, $(C, m) = (0.51, 0.50)$ for $40 \leq \text{Re}_D \leq 10^3$, $n = 0.37$ for $Pr \geq 0.7$ and $n = 0.36$ for $Pr < 0.7$, $Pr/Pr_s \approx 1$, $Pr_{air} \approx 0.71$, and $Pr_{argon} \approx 0.55$.

Case	Location	h ($\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$)	L_c (m)	Nu	Ra or Re
Raised printhead stationary platform (air)	S2	12	$L=0.13$	40	$Ra_L = 3.42 \times 10^7$
	S3	14		47	$Ra_L = 6.58 \times 10^7$
Sidewall buoyancy for raised printhead moving platform (air)	S2	14	$L=0.13$	47	$Ra_L = 6.58 \times 10^7$
	S3	16		53	$Ra_L = 1.07 \times 10^8$

Case	Location	h ($\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$)	L_c (m)	Nu	Ra or Re
Cross-flow for raised printhead moving platform (air)	S2	14	$D=0.0254$	9.1	$Re_D=4.16\times 10^2$
	S3	16		10.4	$Re_D=5.44\times 10^2$
Lowered printhead stationary platform (argon)	S2	9	$L=0.13$	31	N/A
	S3	10		34	
Lowered printhead moving platform (argon)	S2, S3	14	$L=0.13$	48	N/A
Cross-flow for lowered printhead moving platform (argon)	S2, S3	14	$D=0.0254$	9.4	$Re_D=5.18\times 10^2$

Data availability

Data will be made available on request.

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